



ConquerHSC Discord Community

2020

Practise Trial
HSC
Examination

Mathematics – Practise #1

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black pen.
- NESA approved calculators may be used.
- A reference sheet is provided.
- Write your candidate number and/or name on each answer booklet and on your multiple-choice answer sheet.
- Place all your booklets inside the answer booklet for Question Eleven.

Total Marks - 100

Section I – 10 Marks

- Questions 1-10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11-16 are of equal value.
- All necessary working should be shown.
- Write your answers with the relevant working out on the answer sheet provided.

(React with a STONKS emoji if you likey!)

Name:

.....

Teacher:

.....

THIS IS AN UPDATED 2017 TRIAL PAPER!

It does not reflect the format or the content of the 2020 HSC Examination Paper in this subject. In fact, the format of this exam is guaranteed to match the old syllabus HSC exams, NOT the 2020 format. An attempt has been made to update old syllabus questions to new syllabus questions.

Section I

10 marks

Attempt Questions 1 to 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 10

1. What is 7.19632 correct to three significant figures?

(A) 7.196

(B) 7.197

(C) 7.19

(D) 7.20

2. What is the slope of the line with equation $3x - 12y + 5 = 0$?

(A) -4

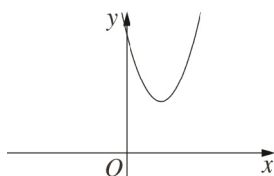
(B) 4

(C) $-\frac{1}{4}$

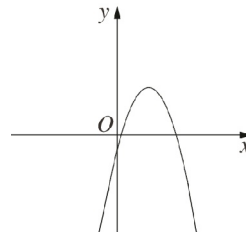
(D) $\frac{1}{4}$

3. Which diagram best shows the graph of the parabola $y = (x + 3)^2 - 4$?

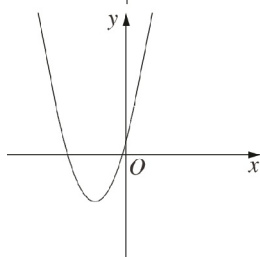
(A)



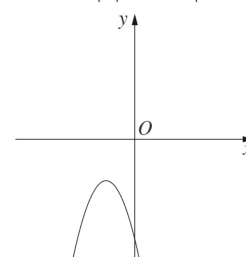
(B)



(C)



(D)



4. Which inequality defines the domain of the function $y = \frac{1}{\sqrt{5+x}}$?

(A) $x > -5$

(B) $x \geq -5$

(C) $x < -5$

(D) $x \leq -5$

5. Let $f(x) = 2 \cos\left(\frac{3x}{\pi}\right)$. What is the period of f ?

(A) $\frac{2}{3}$

(B) 2

(C) $\frac{2\pi}{3}$

(D) $\frac{2\pi^2}{3}$

6. What are the solutions of $2 \cos x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$?

(A) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

(B) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$

(C) $\frac{\pi}{6}$ and $\frac{11\pi}{6}$

(D) $\frac{4\pi}{3}$ and $\frac{5\pi}{3}$

7. What is the derivative of $e^{-\sin x}$?

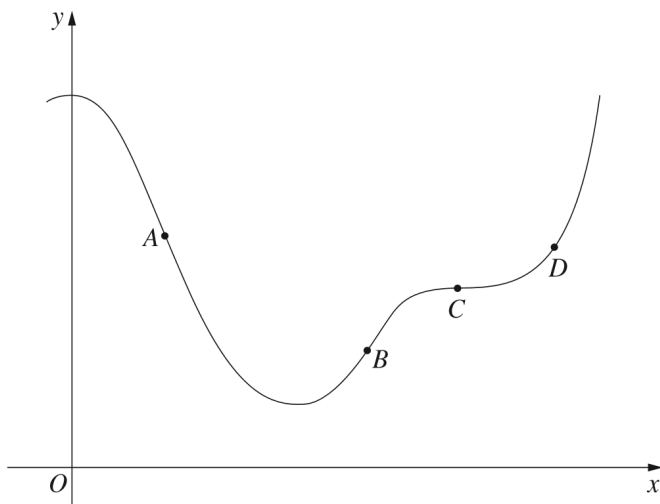
(A) $-\sin x \cdot e^{\cos x}$

(B) $-\cos x \cdot e^{\cos x}$

(C) $-\cos x \cdot e^{-\sin x}$

(D) $\sin x \cdot e^{-\cos x}$

8. The diagram shows points A , B , C and D on the graph $y = f(x)$.



At which point is $f'(x) < 0$ and $f''(x) = 0$?

- (A) A
- (B) B
- (C) C
- (D) D
9. What is the value of $\int_1^9 \frac{1}{2x} dx$?

- (A) $\ln 3$
- (B) $\ln 9$
- (C) $\ln 18$
- (D) $\ln 81$

10. What is the value of $\int_{-1}^2 |x-1| dx$?

- (A) 5
- (B) 1
- (C) $\frac{5}{2}$
- (D) $\frac{3}{2}$

Section II

90 marks

Attempt Questions 11 to 16

Allow about 2hrs 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

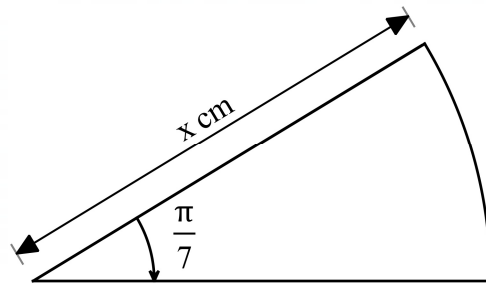
- | | | |
|------------|--|----------|
| (a) | Simplify $5x - (7 - 2x)$ | 1 |
| (b) | Let $f(x) = -x^2 + 3x + 2$ over the domain $[0, \infty)$. Over what interval is the graph of f strictly increasing? | 2 |
| (c) | Express $\frac{7}{3+\sqrt{5}}$ with a rational denominator | 2 |
| (d) | Differentiate $\frac{x^4}{x^2+1}$ | 2 |
| (e) | Differentiate $x^2 \cos x$ | 2 |
| (f) | Let $f(x) = -x^2 + 3x + 2$, and $h(x) = x $. If $g(x) = f(h(x))$, state $g(x)$. | 2 |
| (g) | Find $\int \sin(3x + 2) \, dx$ | 2 |
| (h) | Find $\int \frac{1}{(x-7)^2} dx$ | 2 |

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) The angle of a sector of a circle of radius x cm is $\frac{\pi}{7}$ radians, as shown in the diagram. If the area of the sector is 3 cm^2 , find the exact value of x .

2



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- (b) The cubic $y = 2x^3 + bx^2 + cx + d$ has a point of inflexion at $x = -3$. Find b
- (c) Write down the equation of the new curve when $f(x) = 3x + 2$ is translated left by 3 units.
- (d) Solve $\sin(x - 250^\circ) = \frac{\sqrt{3}}{2}$, where $0^\circ \leq x \leq 360^\circ$.
- (e) Draw a scatterplot that shows low positive correlation.
- (f) Complete the following two-way table.

2

2

2

2

2

	A	$\bar{A}$	
B		3	6
$\bar{B}$			
		4	11

- (g) (i) Differentiate $y = x^2 e^{5x}$
- (ii) Hence find the exact value of $\int_1^4 x e^{5x} (2 + 5x) dx$

1

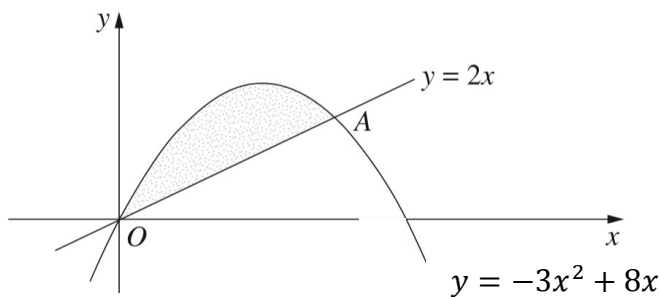
2

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Consider the curve $y = x^3 + x^2 - x - 3$
- (i) Find the stationary points, and determine their nature 4
- (ii) Given that the point P $\left(-\frac{1}{3}, -\frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflexion at P. 2
- (iii) Sketch the curve, labelling the stationary points, point of inflexion and y-intercept. 2

- (b) The parabola $y = -3x^2 + 8x$ and the line $y=2x$ intersect at the origin and the point A.

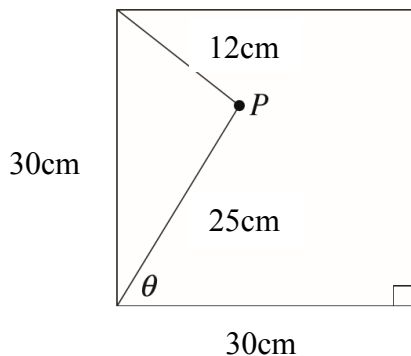


- (i) Find the x -coordinate of the point A 1
- (ii) Calculate the area enclosed by the parabola and the line 3
- (c) Square tiles of side length 30 cm are being used to tile a bathroom. 3

The tiler needs to drill a hole in one of the tiles at a point P which is 12 cm from one corner and 25 cm from an adjacent corner.

To locate the point P the tiler needs to know the size of the angle θ shown in the diagram.

Find the size of the angle θ to the nearest degree.



End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

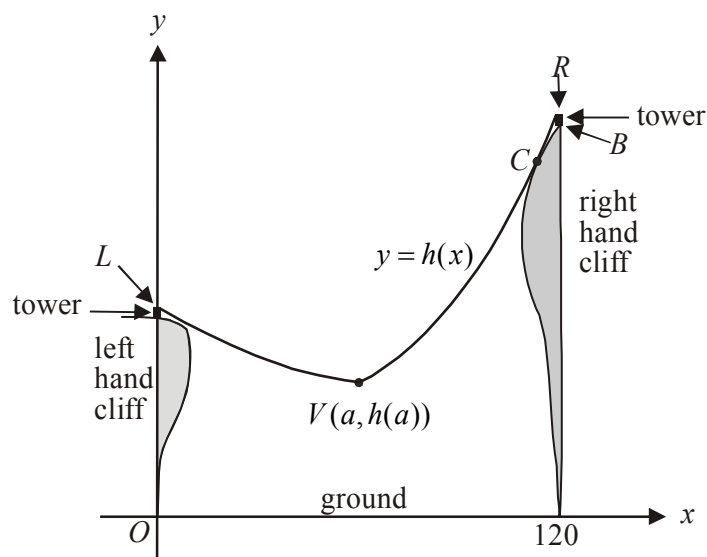
Victoria James is a spy.

Whilst on a chairlift that spans a valley between two cliffs, her enemies destroyed the control tower causing the chairlift cable to slacken.

As a result, the position of the cable above the ground can be described by the **continuous function**

$$h(x) = \begin{cases} \frac{1}{100}(x-50)^2 + 40, & \text{for } x \in [0, a] \\ \frac{1}{100}(x-30)^2 + 40, & \text{for } x \in (a, 120] \end{cases}$$

where x represents the horizontal distance in metres of the cable from the base of the left-hand cliff and y represents the height in metres of the cable above the ground.



The graph of h is shown above.

The cable is attached to the top of 5m high towers on the left- and right-hand cliffs at points L and R respectively. The base of the tower on the right-hand cliff is indicated by point B .

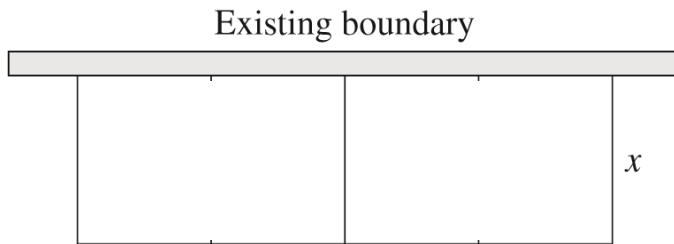
The cable touches the right-hand cliff at point C and starts to fray. Victoria is left stationary at the point $V(a, h(a))$.

- | | | |
|-----|---|---|
| (a) | Find the value of a . | 2 |
| (b) | How high is Victoria above the ground? | 2 |
| (c) | The gradient of the cable at point C is 1.48. Find the coordinates of point C . | 2 |

Victoria has equipment with her that enables her to lower herself to the ground from her stationary position. Let v be her height, in metres, above the ground t seconds after she starts her descent. The rate at which she descends is given by

$$\frac{dv}{dt} = -\frac{23}{21}e^{\left(-\frac{t}{42}\right)}$$

- (d) Find an expression for v in terms of t . 2
- (e) Hence find the time that it takes for Victoria to reach the ground. Express your answer correct to 3 decimal places. 2
- (f) A farmer wishes to make a rectangular enclosure of area 810 m^2 . She uses an existing straight boundary as one side of the enclosure. She uses wire fencing for the remaining three sides and also to divide the enclosure into two equal rectangular areas of width x m as shown.



- (i) Show that the total length, l m, of the wire fencing is given by $l = 3x + \frac{810}{x}$ 2
- (ii) Find the minimum length of wire fencing required, showing why this is the minimum length. 3

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) Find all solutions of $2\cos^2 x + \sin x - 2 = 0$, where $0 \leq x \leq 2\pi$ **3**
- (b) The line $y = mx$ is a tangent to the curve $y = e^{\frac{x}{2}}$ at a point P.
- (i) Sketch the line and the curve on one diagram. **1**
- (ii) Find the coordinates of P. **3**
- (iii) Find the value of m. **1**
- (c) (i) Sketch the graph $y = |3x + 6|$ **1**
- (ii) Using the graph from part (i), or otherwise, find all values of b for which the equation $|3x + 6| = 2x + b$ has two solutions **2**
- (d) The random variable X is normally distributed with mean 142 and variance 16. The random variable Z is the standard normal variable.
- (i) If $\Pr(X < 140) = \Pr(Z > a)$, find the value of a . **1**
- Using the result that $\Pr(X < 140) = 0.3$, correct to 1 decimal place, find:
- (ii) $\Pr(X < 144)$ **1**
- (iii) $\Pr(X < 140 | X < 142)$ **2**

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

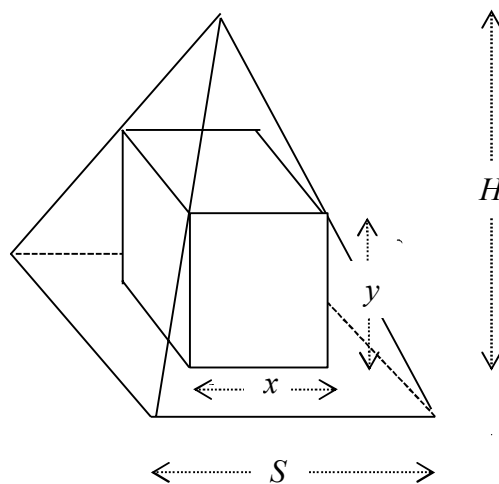
- (a) The derivative of a function $f(x)$ is $f'(x) = 2 \cos x + 7$. The line $y = 8x + \frac{\pi}{3}$ is tangent to the graph of $f(x)$ at the point P.

(i) Find a possible set of coordinates for the point P 3

(ii) Find the function $f(x)$ 2

- (b) The diagram shows a square-based prism of width x and height y inscribed in a square-based pyramid of base width S and height H , where S and H are constants.

The volume V of a square-based pyramid of base s and height h is $V = \frac{1}{3}s^2h$



(i) Show that the volume, V of the prism can be written as $V = \frac{H}{S}x^2(S - x)$ 4

(ii) By considering the inscribed prism of maximum volume, show that the volume of any inscribed prism does not exceed $\frac{4}{9}$ of the volume of the pyramid. 4

(c) Consider the function $y = \sin kx$. 2

Find the smallest value of $k > 0$ so that $\int_0^{\frac{3\pi}{2}} \sin kx \, dx = 0$

End of Paper